

Introduction to Large Language Models (LLMs)

LARGE LANGUAGE MODELS

Chapter 1

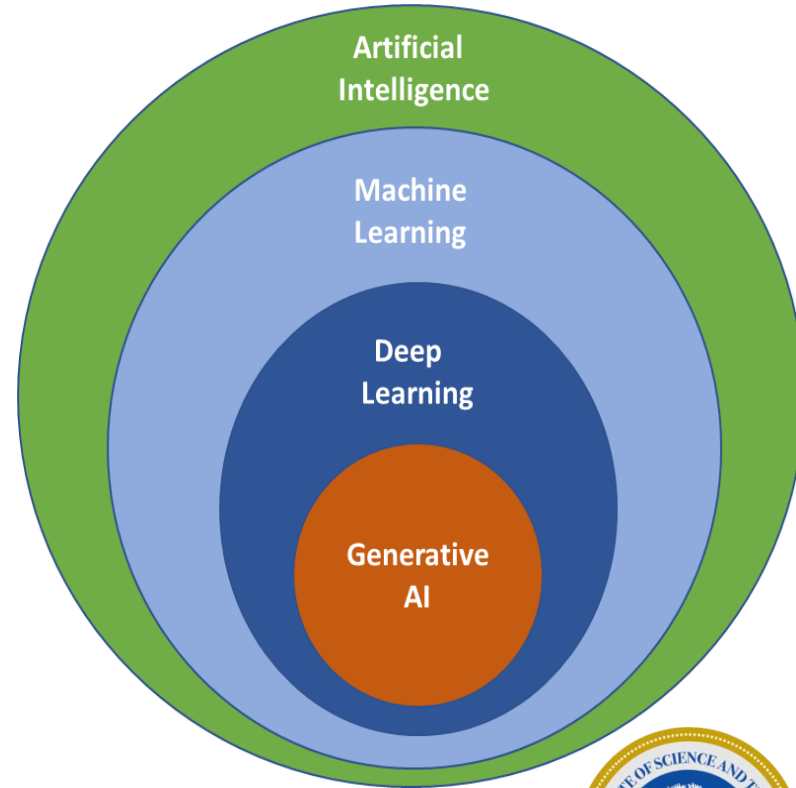


Is this AI?



Generative AI

- **Traditional AI (TA):**
 - Focuses on analyzing existing data, performing specific tasks, and making predictions based on predefined rules and algorithms. It excels at pattern recognition and decision-making within a defined scope.
- **Generative AI (GenAI):**
 - Creates new content, like text, images, or code, by learning patterns from data and generating novel outputs. It's more adaptable and creative, capable of producing outputs not explicitly present in the training data.



Beginning of everything

LANGUAGE MODEL



Language Model

- At its core, a **Language Model (LM)** is a statistical tool that calculates the probability of a sequence of words occurring in a particular order. More simply, it tries to predict the next word in a sequence given the words that have come before it.
- **What it does:**
 - Assigns a probability score to a given sequence of words.
 - Predicts the most likely next word (or words) in a sentence.



Language Model

- "The patient felt much better after taking the prescribed..."

medicine
pills
dosage

*mountain
*television



Why word prediction?

Grammar and Style Correction:

- **User Input:** "The team will meat on Tuesday."
- **LM Correction:** "The team will **meet** on Tuesday."

Predictive Search / Query Expansion:

- **User Input:** "Best rest..."
- **LM Prediction:** "Best **restaurants near me**"



Why word prediction?

Improving Machine Translation:

- **Raw Translation** : "She beautiful house living in."
- **LM Refinement**: "She is living in a beautiful house."

Voice Assistant Understanding:

- **User Spoken Input** : "Set an alarm for ate o'clock."
- **LM Interpretation**: "Set an alarm for **8 o'clock**."
- Many more....



Language Models: A History

Early Beginnings (1940s-1980s):

- Early models were often rule-based, relying on manually crafted grammatical rules. These were rigid and struggled with ambiguity and scale.

The Rise of Statistical Models (1980s-Early 2010s):

- The breakthrough came with the shift towards **statistical methods**. Instead of hand-coding rules, models learned patterns from large text corpora (collections of text).
- **N-gram models** emerged as the dominant approach, leveraging Markov assumptions to simplify probability calculations. These were simpler, more robust, and could be trained on vast amounts of data.
- They were the workhorse for applications like speech recognition throughout this period.



Language Models: A History

The Neural Revolution (Mid-2010s - Present):

- The advent of powerful neural networks (especially Recurrent Neural Networks like LSTMs and later, Transformers) revolutionized LMs.
- Neural LMs can capture much longer-range dependencies and complex linguistic patterns than their statistical predecessors.
- This led to the development of highly sophisticated models like Word2Vec, ELMo, BERT, GPT, and many more, which have dramatically improved performance across nearly all NLP tasks.



Probabilistic Language Models

At their core, all language models are **probabilistic**. They assign a probability to a sequence of words.

Let's consider a sequence of words $W=w_1, w_2, \dots, w_N$.

A probabilistic language model aims to calculate the probability of this entire sequence, $P(W)$ or $P(w_1, w_2, \dots, w_N)$.



Probabilistic Language Models

Goal: compute the probability of a sentence or sequence of words W :

- $P(W) = P(w_1, w_2, w_3, w_4, w_5 \dots w_n)$

Related task: probability of an upcoming word:

- $P(w_5 | w_1, w_2, w_3, w_4)$ or $P(w_n | w_1, w_2 \dots w_{n-1})$

An LM computes either of these:

- $P(W)$ or $P(w_n | w_1, w_2 \dots w_{n-1})$



Probabilistic Language Models

Could we just count and divide?

$P(\text{sky} | \text{The clouds drifted across the})$

- If we tried to calculate this directly by counting:

$C(\text{The clouds drifted across the})$

$C(\text{The clouds drifted across the sky})$

- **No! Still too many unique combinations!**
- Even for a five-word sequence like "The clouds drifted across the sky," the likelihood of seeing this *exact sequence* repeatedly in any given training corpus is very low.



How to compute $P(W)$ or $P(w_n | w_1, \dots, w_{n-1})$

How to compute the joint probability $P(W)$:

$P(\text{The, cloud, drifted, sky, across})$

Intuition: let's rely on the Chain Rule of Probability



Reminder: The Chain Rule

- Recall the definition of conditional probabilities
 - $P(B|A) = P(A,B)/P(A)$ Rewriting: $P(A,B) = P(A) P(B|A)$
- More variables:
 - $P(A,B,C,D) = P(A) P(B|A) P(C|A,B) P(D|A,B,C)$
- The Chain Rule in General
 - $P(x_1, x_2, x_3, \dots, x_n) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2) \dots P(x_n|x_1, \dots, x_{n-1})$



Markov Assumption

The probability of a word appearing depends only on a fixed, limited number of preceding words, rather than the entire preceding sequence.

$P(\text{sky}|\text{The clouds drifted across the})$

$\approx P(\text{sky}|\text{the})$

$$P(w_{n+1}|w_n) \approx P(w_n|w_{n-1})$$



Markov Assumption

For an **N-gram model**, this means the probability of the current word (w_i) depends only on the previous **N-1** words.

For a **Bigram Model (N=2)**: The probability of w_n depends only on w_{n-1} .

$$P(w_n | w_{n-1}) \approx P(w_n | w_{n-1})$$

For a **Trigram Model (N=3)**: The probability of w_n depends only on w_{n-2} and w_{n-1} .

$$P(w_n | w_{n-1}) \approx P(w_n | w_{n-2}, w_{n-1})$$



Limitations of N Gram

N-grams can't handle long-distance dependencies

"The **decision** to implement these drastic policy changes, which faced strong opposition from various stakeholders, **was** ultimately made by the committee."

- Here, the singular verb "**was**" needs to agree with the singular subject "**decision**."
- An N-gram model, with its limited window, might see "stakeholders was" or "committee was" and struggle to correctly predict "was" because the subject "decision" is many words away.



Limitations of N Gram

N-grams don't do well at modeling new sequences with similar meanings

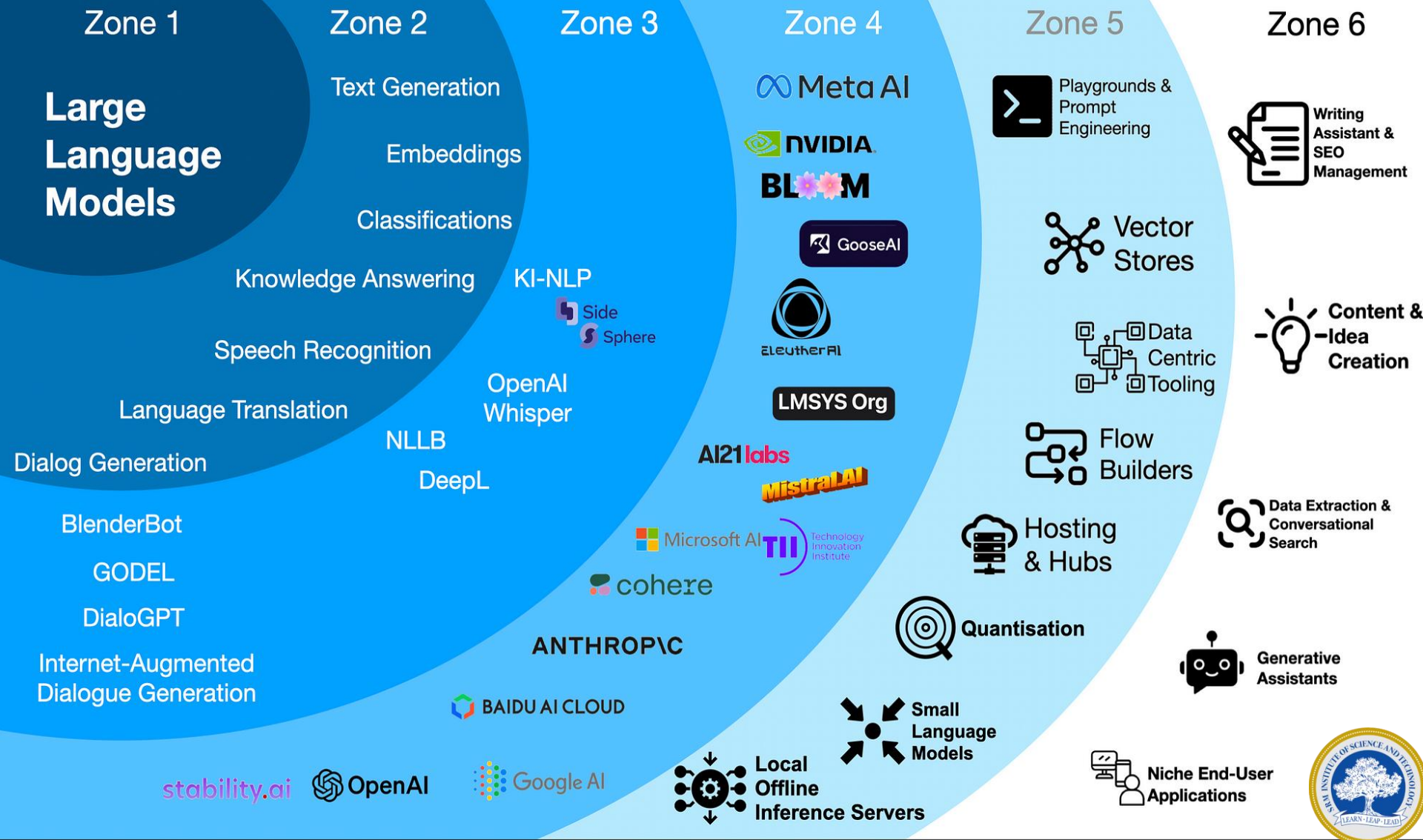
- If an N-gram model has often seen "The **student** understood the lesson," it will assign a high probability to "student understood."
- However, if it has *never* encountered "The **pupil** grasped the concept" in its training data, it might assign a very low or zero probability to "pupil grasped," even though "student" and "pupil" are synonyms, and "understood" and "grasped" are similar in context.



Large Language Models (LLMs)

- Large Language Models (like BERT, GPT, Gemini) were developed to overcome these fundamental limitations of N-gram models, primarily due to their underlying architecture (e.g., the Transformer model) and the way they represent words.
- It Can
 - Handle Much Longer Contexts:
 - Model Synonymy & Generate Better Novel Strings





Large Language Models (LLMs)

- **Popular LLMs Today**

- GPT-4o (OpenAI), Claude 3.5 (Anthropic), Gemini 1.5 (Google).

- **Capabilities**

- Human-like text generation, content creation, coding, and reasoning.

- **Advancements**

- Better context understanding, memory, and efficiency.



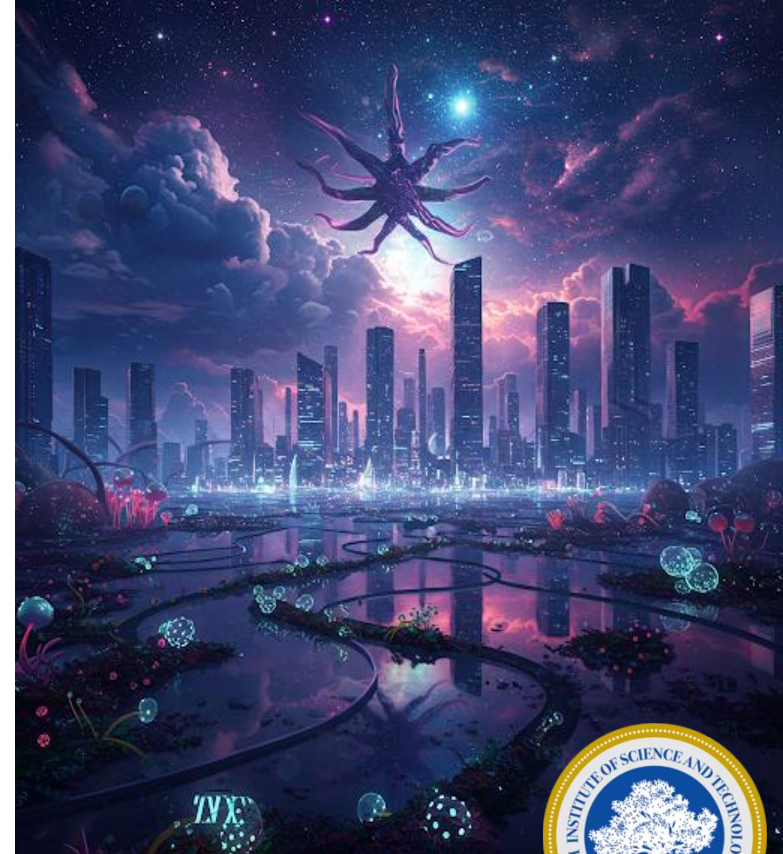
Multimodal AI (Text, Image, Video, Audio)

- **What is Multimodal AI?**
AI that processes multiple data types (text, images, video, speech).
- **Latest Developments GPT-5V** (vision capabilities).
- **Gemini 3.5** (strong multimodal processing).
- **Sora 2**(AI video generation).



AI-Generated Images & Art

- **Popular Tools**
 - Midjourney, DALL·E 3, Stable Diffusion.
- **Use Cases**
 - Digital art, marketing, gaming, design, and advertising.
- **Advancements**
 - Higher resolution, more control, and better consistency in outputs.



AI-Generated Video

- **Sora / Veo**– Creates high-quality videos from text prompts.
- **Runway AI** – Used in filmmaking and content creation.
- **Applications** Film production, advertising, animation, and education.



AI in Music & Audio

- **AI Music Generation Tools Suno, Stable Audio, AIWA, Boomy.**
- **How AI is Changing Music?**
 - AI-assisted composition, remixing, and sound enhancement.
- **Impact**
 - Revolutionizing background music for videos, games, and creative work.



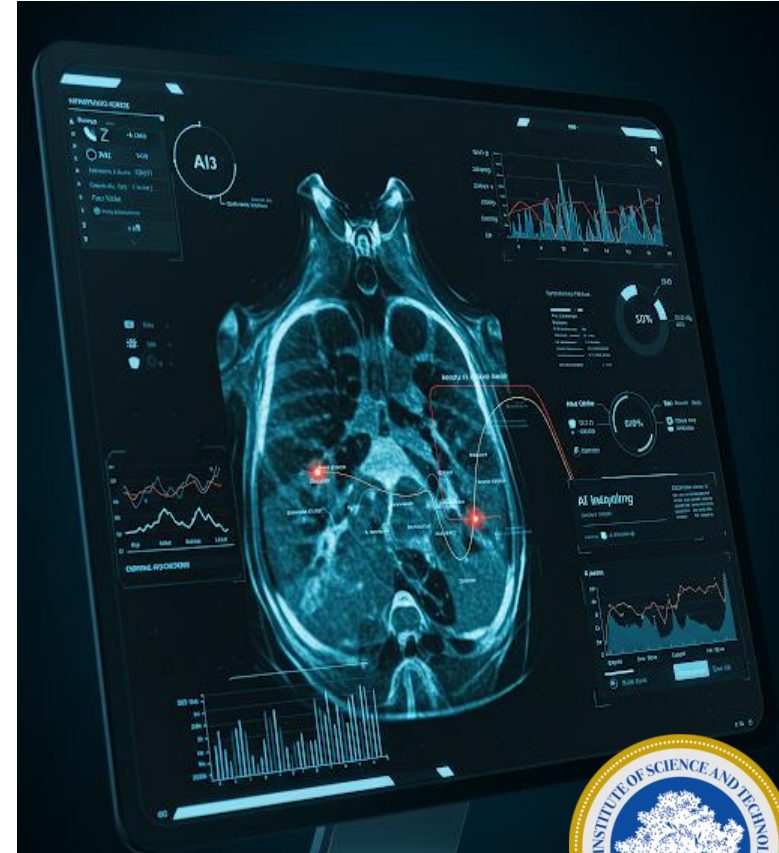
AI in Gaming & Virtual Worlds

- AI-powered NPCs with dynamic interactions.
- AI-assisted game level and environment design.
- AI-generated textures, animations, and game scripts.
- **Examples**
 - **Unreal Engine's MetaHumans** (realistic character creation).
 - AI-generated narratives in RPGs.



AI in Healthcare

- **AI Applications in Medicine**
 - AI-assisted **drug discovery** (reducing development time).
 - AI-enhanced **medical imaging** (faster & more accurate diagnosis).
 - AI-powered **personalized treatment plans**.
- **Recent Advancements**
 - AI models predicting diseases more accurately than doctors in some cases.





AI AGENTS TAKES THE SHOW



Challenges & Ethical Concerns

- **Major Concerns Misinformation & Deepfakes:**
 - AI-generated fake news and altered media.
- **Bias in AI Models:**
 - AI reflecting societal biases.
- **Data Privacy Issues:**
 - AI using personal data.
- **Copyright & Ownership:**
 - Who owns AI-generated content?
- **Regulatory Actions**
 - AI laws and guidelines being developed globally.



Future of Generative AI

- **What's Next?**

- More **personalized AI assistants**.
- **AI reasoning & decision-making improvements**.
- AI helping with **scientific discoveries**.

- **Long-Term Impact**

- AI becoming an essential part of daily life, work, and research.



Prompts

Prompt



Input



Generated Text



Output



LeewayHertz



Prompt Engineering

- At its heart, prompt engineering is akin to teaching a child through questions. Just as a well-phrased question can guide a child's thought process, a well-crafted prompt can steer an AI model, especially a Large Language Model (LLM), towards a specific output.
- Prompt engineering is the bridge ensuring effective human-AI communication. It's not just about getting the right answer; it's about ensuring AI understands the context, the nuances and the intent behind every query.



Art of Prompt engineering- 101

- Start by clearly defining your objective, then be as specific as possible, providing detailed information and examples where relevant, breaking down complex tasks into smaller parts, and iteratively refining your prompts based on the AI's responses, constantly checking for clarity and adjusting as needed.
- Essentially, treat prompt engineering as a conversation where you guide the AI towards the desired output through precise instructions.



Be Specific as possible

- **Detailed context:** Provide the AI with enough background information to understand the scenario you're inquiring about. This includes the subject matter, scope, and any relevant constraints.
- **Desired format:** Clearly specify the format in which you want the information to be presented, whether it's a list, a detailed report, bullet points, or a summary. Mention any structural preferences, such as headings, subheadings, or paragraph limits.
- **Output length:** Detail how long you want the AI's response, whether "3 paragraphs" or "250 words."
- **Level of detail:** Indicate the level of detail required for the response, from high-level overviews to in-depth analysis, to ensure the model's output matches your informational needs.
- **Tone and style:** Request the preferred tone and style, whether it's formal, conversational, persuasive, or informational, to make sure the output aligns with your intended audience or purpose.
- **Examples and comparisons:** Ask the AI to include examples, analogies, or comparisons to clarify complex concepts or make the information more relatable and easily understood.



Prompt Engineering 101

- Supply the AI with examples
- Specify your desired output
- Provide instructions on what to do instead of what not to do
- Give the model a persona or frame of reference
- Try chain of thought prompting
- Split complex tasks into simpler ones



Example

- Please provide an outline for a comprehensive report that analyzes the current trends in social media marketing for technology companies, focusing on the developments from 2020 onward.
- The outline should include an introduction, three main sections addressing different aspects of social media trends, and a conclusion summarizing the findings. Please suggest the types of graphs that could illustrate user engagement trends and list bullet points that summarize key marketing strategies in each section.



Example

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅



Example

- I went to the market and bought 10 apples. I gave 2 apples to the neighbor and 2 to the repairman. I then went and bought 5 more apples and ate 1. How many apples did I remain with?
- I went to the market and bought 10 apples. I gave 2 apples to the neighbor and 2 to the repairman. I then went and bought 5 more apples and ate 1. How many apples did I remain with? Let's think step by step.



Prompt Engineering

- **Goal:** Create a short story about a robot experiencing human emotions, specifically romantic feelings for a human character.
- "Write a story about a robot who unexpectedly falls in love with a human."



Prompt Engineering

- **Goal:** Create a short story about a robot experiencing human emotions, specifically romantic feelings for a human character.
- "Write a short story titled 'Circuitry of the Heart' where a highly advanced domestic robot named 'AX-7' begins to feel a deep connection with its human owner, a young programmer named Sarah, while noticing subtle human behaviors and emotions in her daily routine. Describe the internal conflict AX-7 experiences as its robotic programming clashes with these emerging feelings."



Challenges faces and Methods used

MODEL EVALUATION



Need for Model Evaluation

Model Trained



Expectation

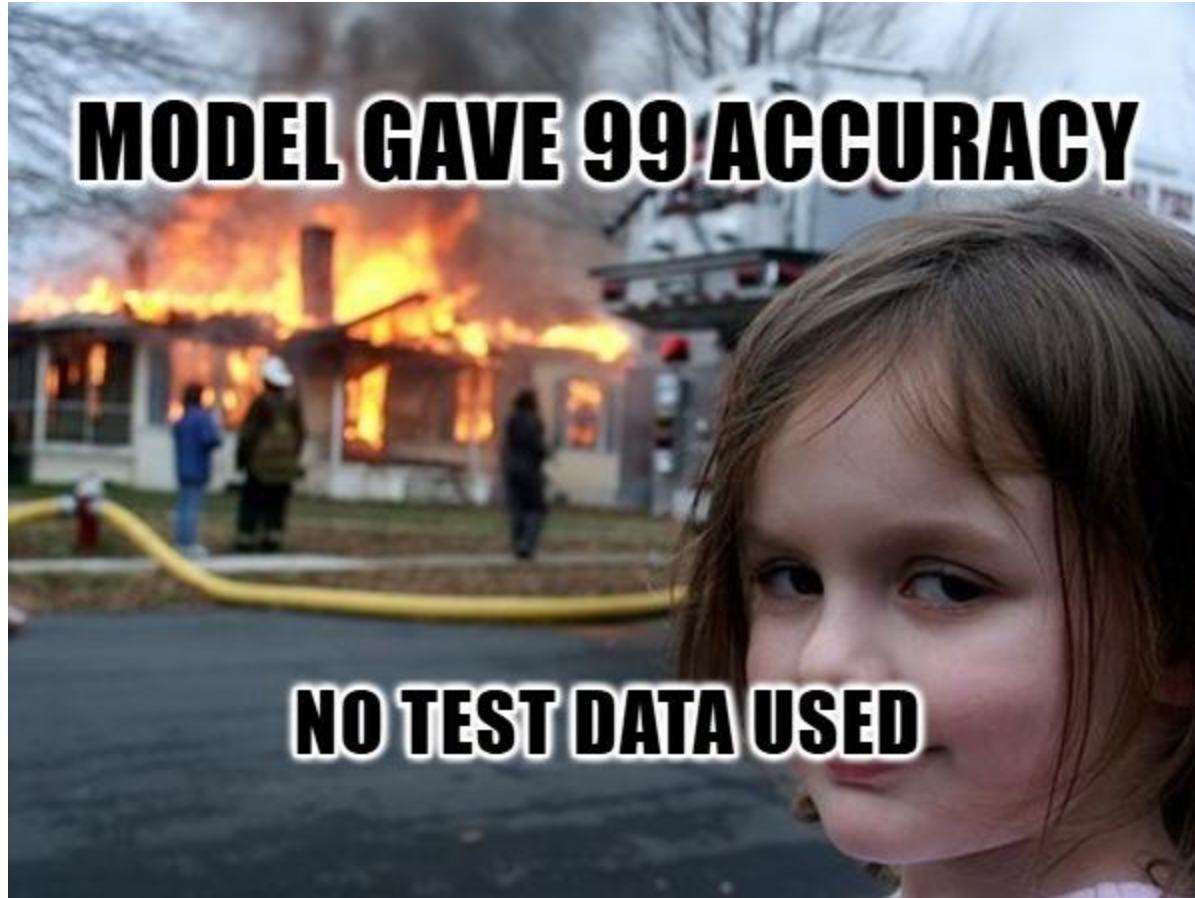
Model Deployed



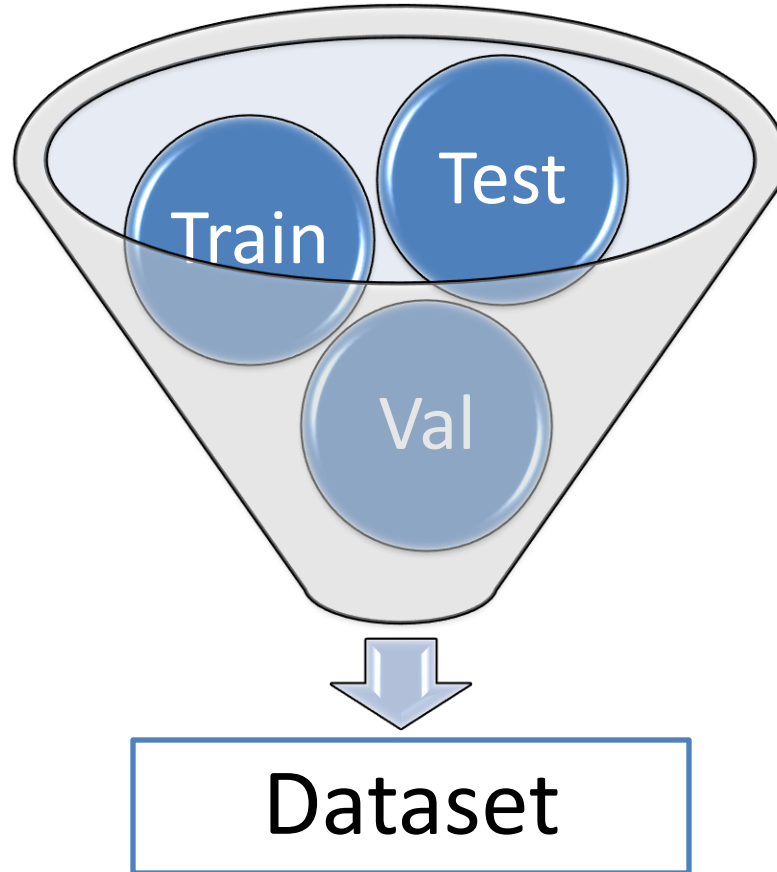
Reality



Need for Model Evaluation

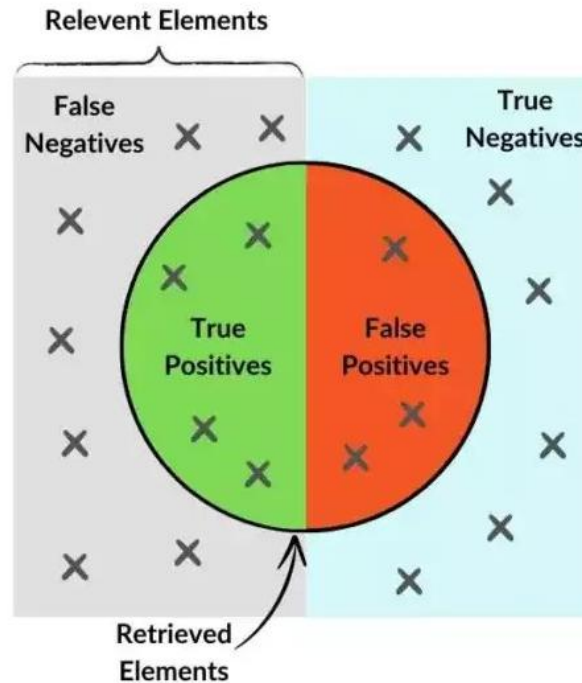


Train-Test Split



Standard Metrics

Data	Ground Truth	Prediction
Data 1	Class A	Class B
Data 2	Class B	Class B
Data 3	Class A	Class A
Data 4	Class B	Class A



How many retrieved elements are relevant?

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

How many relevant elements are retrieved?

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$



Standard Metrics

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

$$\text{Accuracy} = \frac{\text{I love this class}}{\text{I love this class}}$$



Challenges Faced

Source	The Bells of St. Martin's Fall Silent as Churches in Harlem Struggle .
Translation	Die Glocken von St. Martin verstummen , da Kirchen in Harlem Probleme haben .
Paraphrase	Die Probleme in Harlems Kirchen lassen die Glocken von St. Martin verstummen .
Paraphrase	Die Kirchen in Harlem kämpfen mit Problemen , und so läuten die Glocken von St. Martin nicht mehr .

Freitag et al. (2020). BLEU might be Guilty but References are not Innocent

There can be many semantically equivalent but syntactically different ways of expressing the same idea.



Challenges Faced

Semantic Equivalence

Deep really loves Dessert = Deep adores Dessert



Semantic Contrast

He did **not** pass the exam $\neq$ He did pass the exam



Model Evaluation – Standard methods

- Perplexity
- ROUGE (Recall-Oriented Understudy for Gisting Evaluation)
- BLEU (Bilingual Evaluation Understudy)



Perplexity



Perplexity



Don't confuse it with ChatGPT's "rival" – we're talking about Perplexity!

Perplexity is a crucial metric used to evaluate how effectively a language model predicts a sequence of words. What's particularly useful is that it **doesn't require ground truth** – making it a versatile and powerful tool.



Perplexity

"The sky is blue, the grass is green, and the sun is a bright _____"

"Twinkle, twinkle, little _____"

In essence, Perplexity measures how "Perplexed / Surprised" a language model is by new data.

A model that is less surprised is a better model.



Perplexity

- A perplexity score of 1 indicates that the model predicted the sequence with perfect accuracy.
- Conversely, a perplexity of 7 suggests that the model, on average, considered 7 equally probable alternatives at each prediction point, signifying a higher degree of uncertainty.

$$\exp \left(-\frac{1}{N} \sum_{i=1}^N \log P(w_i | w_1, w_2, \dots, w_{i-1}) \right)$$



Perplexity

- Perplexity is fundamentally related to the **probability** a language model assigns to a sequence of words.
- The general formula for perplexity (PP) of a sequence of words $W=w_1, w_2, \dots, w_N$ is:

$$PP(W) = P(w_1, w_2, \dots, w_N)^{-\frac{1}{N}}$$

Where:

- $P(w_1, w_2, \dots, w_N)$ is the **probability** assigned by the language model to the entire sequence of N words.
- N is the **number of words** in the sequence.



Perplexity

Sentence: "The cat sat" (So, $N=3$ words)

**Assumed Language Model Probabilities
(Hypothetical):**

- $P(\text{The}) = 0.05$
- $P(\text{cat} \mid \text{The}) = 0.40$
- $P(\text{sat} \mid \text{The cat}) = 0.70$



Perplexity

Step 1 - Calculate the logarithm

For w_1 ="The": $\log P(\text{The}) = \log(0.05) \approx -2.9957$

For w_2 ="cat": $\log P(\text{cat} \mid \text{The}) = \log(0.40) \approx -0.9163$

For w_3 ="sat": $\log P(\text{sat} \mid \text{The cat}) = \log(0.70) \approx -0.3567$

Step 2 - Sum the log probabilities

Sum $= (-2.9957) + (-0.9163) + (-0.3567) \approx -4.2687$



Perplexity

Step 3 - Calculate the logarithm

$$-\frac{1}{3} \times (-4.2687) \approx 1.4229$$

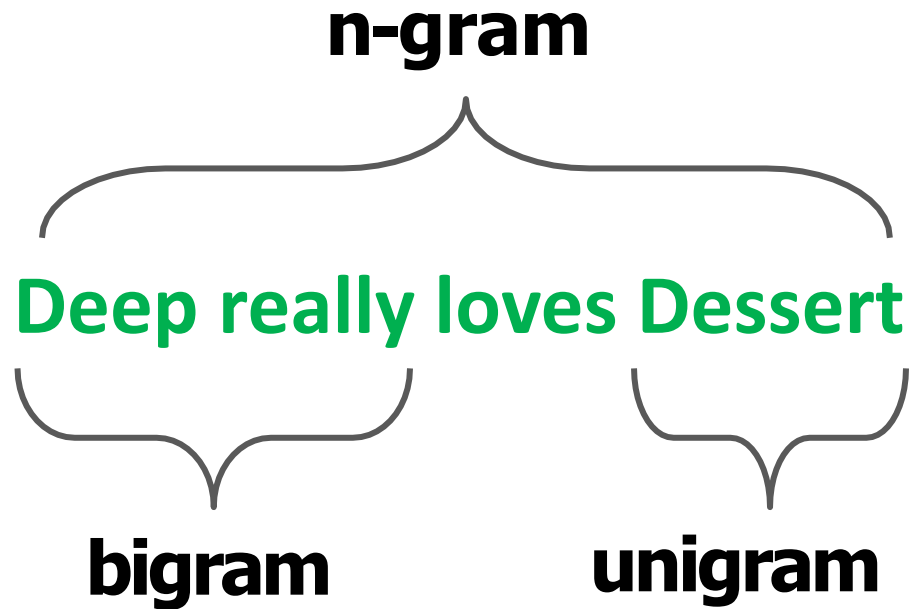
Step 4 - Apply the exponential function

$$\text{Perplexity} = \exp(1.4229)$$

$$\text{Perplexity} \approx 4.148$$



LLM Evaluation-Metrics-Terminology



ROUGE

- Recall-Oriented Understudy for Gisting Evaluation
- ROUGE-1
- ROUGE-2
- ROUGE-L



ROUGE-1

Reference (human):

This dish is good

Generated output:

This dish is very good

$$\text{ROUGE-1 Recall} = \frac{\text{unigram matches}}{\text{unigrams in reference}} = \frac{4}{4} = 1.0$$

$$\text{ROUGE-1 Precision:} = \frac{\text{unigram matches}}{\text{unigrams in output}} = \frac{4}{5} = 0.8$$

$$\text{ROUGE-1 F1:} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = 2 \frac{0.8}{1.8} = 0.89$$



ROUGE-1

Reference (human):

This dish is good

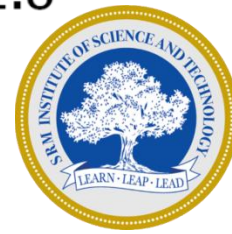
Generated output:

This dish is not good

$$\text{ROUGE-1 Recall} = \frac{\text{unigram matches}}{\text{unigrams in reference}} = \frac{4}{4} = 1.0$$

$$\text{ROUGE-1 Precision:} = \frac{\text{unigram matches}}{\text{unigrams in output}} = \frac{4}{5} = 0.8$$

$$\text{ROUGE-1 F1:} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = 2 \frac{0.8}{1.8} = 0.89$$



ROUGE-2

Reference (human):

This dish is good

This dish

dish is

is good

Generated output:

This dish is very good

This dish

dish is

is very

very good



ROUGE-2

Reference (human):

This dish is good

This dish

dish is

is good

Generated output:

This dish is very good

This dish

dish is

is very

very good

$$\text{ROUGE-2 Recall:} = \frac{\text{bigram matches}}{\text{bigrams in reference}} = \frac{2}{3} = 0.67$$

$$\text{ROUGE-2 Precision:} = \frac{\text{bigram matches}}{\text{bigrams in output}} = \frac{2}{4} = 0.5$$

$$\text{ROUGE-2 F1:} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = 2 \frac{0.335}{1.17} = 0.57$$



ROUGE-L

Reference (human):

This dish is good

Generated output:

This dish is very good

Longest common subsequence (LCS):

This dish



ROUGE-L

Reference (human):

He feels very well

Generated output:

He feels quite very well

Longest common subsequence (LCS):

He feels

very well



ROUGE-L

Reference (human):

He feels very well

Generated output:

He feels quite very well

$$\text{ROUGE-L Recall:} = \frac{\text{LCS}(\text{Gen}, \text{Ref})}{\text{unigrams in reference}} = \frac{2}{4} = 0.5$$

$$\text{ROUGE-L Precision:} = \frac{\text{LCS}(\text{Gen}, \text{Ref})}{\text{unigrams in output}} = \frac{2}{5} = 0.4$$

$$\text{ROUGE-L F1:} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = 2 \frac{0.2}{0.9} = 0.44$$



ROUGE hacking

Reference (human):

He feels very well

Generated output:

well well well well



ROUGE hacking

Reference (human):

He feels very well

Generated output:

well well well well

$$\text{ROUGE-1 Precision} = \frac{\text{unigram matches}}{\text{unigrams in output}} = \frac{4}{4}$$

$$\text{Modified precision} = \frac{\text{clip}(\text{unigram matches})}{\text{unigrams in output}} = \frac{1}{4}$$



BLEU

- **BLEU** stands for **Bi**Lingual **E**valuation **U**nderstudy.
- It is a widely used automatic metric for evaluating the quality of machine-translated text.
- Its core idea is to measure the similarity between a machine-generated translation (the "candidate") and one or more high-quality human translations (the "references").



BLEU

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log P_n \right)$$

Where:

- BP is the Brevity Penalty.
- N is the maximum n-gram size considered (e.g., 4 for BLEU-4).
- w_n are positive weights for each n-gram precision (typically $w_n=1/N$ for uniform weighting, e.g., 0.25 for BLEU-4).
- P_n is the clipped n-gram precision for n-gram length 'n'.



BLEU

- N-gram Precision (P_n):

$$P_n = \frac{\sum_{\text{n-gram} \in \text{Candidate}} \text{Count}_{\text{clip}}(\text{n-gram})}{\sum_{\text{n-gram} \in \text{Candidate}} \text{Count}(\text{n-gram})}$$

- Brevity Penalty (BP):

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

c is the length of the candidate translation.
 r is the effective reference length



BLEU

BLEU Metric = Avg(precision across range of n-gram sizes) * Brevity Penalty

Reference (human): "The capital city of France is Paris, a beautiful city."

Generated outputs and their (illustrative) BLEU Scores:

- "The capital of France is beautiful." - **BLEU 0.350**
(Missing "city," "Paris," "city." Significant information loss.)
- "The capital city of France is Paris." - **BLEU 0.680**
(Closer, but misses the descriptive "a beautiful city.")
- "The capital city of France is Paris, a beautiful." - **BLEU 0.850**
(Very close, only missing the final "city.")
- "The capital city of France is Paris, a beautiful city." - **BLEU 1.000**
(Perfect match to the reference.)



Challenges Faced

Semantic Equivalence

Deep really loves Dessert = Deep adores Dessert



Semantic Contrast

He did **indeed** pass the exam $\neq$ He did pass the exam



Need a model for Evaluation



End of Slide

THANK YOU

